

Matthew Moreno

Miami, FL | (305) 250-8926 | morenomatthew10@yahoo.com

OBJECTIVE

Electrical and Computer Engineering student seeking an Engineering Internship where I can apply hands-on skills in electronics, digital systems, FPGA design, and prototyping. Driven by curiosity and a strong self-taught mindset, I am passionate about robotics, tinkering, and building practical solutions, and motivated by creating useful technologies that help people and solve real-world problems through thoughtful engineering.

EDUCATION

Miami Dade College — Miami, FL

August 2022 – Present

Bachelor of Science in Electrical & Computer Engineering

- **Relevant Coursework:** Digital Systems & Logic Design, Computer Engineering Fundamentals, Electronics & Circuit Analysis
- **Certifications:** CompTIA A+ IT; Photovoltaic and Design Installation Certified

TECHNICAL SKILLS

- **Programming & HDL:** C, VHDL, Arduino (Embedded Programming), Raspberry Pi
- **Digital & Electrical Engineering:** FPGA Design, Combinational & Sequential Logic, Finite State Machines (FSMs), Truth Tables, Karnaugh Maps, Timing Analysis, Binary Counters
- **Electronics & Hardware:** Circuit Prototyping, Soldering, AC/DC Power Conversion, Oscilloscope, Digital Multimeter, Hardware Debugging & Testing
- **3D Design & Prototyping:** 3D Modeling, 3D Printing (FDM), Design Iteration
- **Engineering Tools:** Vivado, Vitis, Microchip Studio, VirtualBox, Microsoft Office Suite
- **Operating Systems:** Windows, Linux (installation, configuration, troubleshooting)

ENGINEERING PROJECTS

AC-to-DC Voltage Converter

- Designed and implemented a functional AC-to-DC power supply applying rectification, filtering, and voltage regulation techniques.
- Verified output stability using oscilloscopes and multimeters.

FPGA-Based Vending Machine Controller

- Developed a vending machine controller using FPGA digital logic.
- Implemented finite state machines (FSMs) (states, inputs, transitions).
- Built combinational and sequential circuits (logic gates).
- Validated designs using Vivado, Vitis, and Microchip Studio.

Digital 12-Hour Clock

- Designed a digital clock using sequential logic and timing circuits.
- Implemented binary counters and clock division logic (flip-flops, clock signals).
- Applied core digital systems concepts including synchronization and state control.

TECHNICAL & PROFESSIONAL EXPERIENCE

Executive Administrative Intern — City of Doral, Office of the City Council

August 2026 – Present

- Support daily operations of a City Councilmember's office, coordinating schedules, correspondence, and constituent requests. Prepare briefing materials and assist with community events and resident outreach.

Robotics & Coding Instructor — Teknik Labs (Self-Contracted)

July 2025 – Present

- Teach afterschool robotics and coding to students as a self-contracted instructor, designing engaging lessons that build foundational programming and engineering skills. Mentor students through hands-on projects and adapt instruction to varied learning levels.

Solar Sales & Logistics Manager — GC Solar and Electric

March 2023 – April 2025

- Designed custom solar panel layouts considering electrical constraints and system efficiency. Applied technical knowledge to real-world renewable energy systems and supported installation planning.

LEADERSHIP & STRENGTHS

- IEEE Student Officer
- Passion for robotics and hands-on engineering
- Strong problem-solving and analytical skills
- Team collaboration and technical communication
- Detail-oriented, adaptable, and quick learner
- Bilingual (English & Spanish)